Bernoulli Transport Sieves and Finite-Range Integer-Source Triple-Zero Exclusion in Stationary CCR Polynomials

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October 2026

Abstract

We study the impossibility of exact triple zeros in a source-dependent family of stationary chronological-carry-refinement (CCR) polynomials. For every odd order $5 \le n \le 75$, every prime transport parameter $P > 2n$, and all ordinary-integer source vectors, we prove that the polynomial cannot be divisible by X^3. The proof is uniform in the sources and in P: transport-prime reduction forces P to divide a Bernoulli numerator, while prime-order and prime-plus-two congruences supply independent auxiliary-prime obstructions. For exceptional composite orders we give certified finite-ring identities valid for every source choice, including a source-uniform fifth-adic Frobenius-function obstruction at order 65. The result is a finite-order theorem, but not a bounded search over source vectors or transport primes. Exact rational computations, integer-prime certificates, and original-recurrence consistency checks accompany the proof. The statement concerns the specified formal stationary CCR model and does not assert unrestricted machine-realizability conclusions.

Keywords: Bernoulli numbers; p-adic congruences; stationary polynomial transport; integer-source obstruction; Sheffer series; certified computation.
2020 Mathematics Subject Classification: Primary 11B68; Secondary 11A07, 11S05.

1 Introduction and stationary transport

Relation to preceding CCR research. The chronological-carry-refinement transport originates in the author's third-order, fourth-order, and general higher-order CCR studies [1, 2, 3]. Earlier arithmetic refinement classified selected source lattices, local ramification phenomena, and p-adic densities [4]. A related treatment of stationary transports established a sharp valuation law for a specific high-degree coefficient [5], whereas the author's single-site paper addresses prime-indexed Nörlund–Stirling root rigidity [6]. The result here concerns a different question: the *simultaneous* vanishing of the three lowest coefficients for *every* integer-source choice at all odd orders in a prescribed finite interval. We do not claim that classical Bernoulli congruences, p-adic lifting, or finite-field ideal elimination are novel in isolation. Relevant prior arithmetic literature includes [7, 8].

For an integer parameter P and source variables $\lambda_3, \lambda_4, \ldots$, put $F_0(X) = 1$ and $F_j(X) = 0$

for $j < 0$. For $j \geq 1$ define the rational polynomial F_j by

$$\begin{aligned} T_j F_j(X) &= ((P+1)X - P)F_{j-1}(X) \\ &\quad + P\big((P+1)(X-1) - P\big)F_{j-2}(X-1) \\ &\quad + P\sum_{s=3}^{j}(-1)^s \lambda_s F_{j-s}(X-s+1), \qquad (1) \\ T_j h(X) &= Xh(X+1) - (X-j)h(X). \end{aligned}$$

The operator T_j is upper triangular on coefficients in the monomial basis, with nonzero diagonal $j, j+1, \ldots$, so this determines F_j uniquely over $\mathbb{Q}$.

A *triple zero at the origin* means

$$X^3 \mid F_n(X) \quad \Longleftrightarrow \quad [X^0]F_n = [X^1]F_n = [X^2]F_n = 0.$$

Here "integer sources" always means *all* $\lambda_3, \ldots, \lambda_n \in \mathbb{Z}$; it does not refer to arbitrary rational or merely locally integral sources. This distinction is essential: as we recall below, suitable local source vectors often *can* produce triple zeros.

Theorem 1.1 (Certified finite-range exclusion)**.** *For each odd integer n with $5 \leq n \leq 75$, each prime $P > 2n$, and every source vector $(\lambda_3, \ldots, \lambda_n) \in \mathbb{Z}^{n-2}$, the polynomial F_n defined by* (1) *satisfies*

$$X^3 \nmid F_n(X).$$

This theorem is *finite in n* but uniform in the transport prime and in *all* integer sources. The proof combines general source-independent congruences with exact coefficientwise certificates at the exceptional orders. No finite enumeration of integer-source vectors is assumed.

2 Contributions and novelty

This paper makes three specific contributions relative to the preceding CCR source literature. First, it gives an *all-integer-source* nonexistence theorem at each of the 36 odd orders from 5 to 75, for *every* transport prime $P > 2n$; this is stronger than excluding sampled source vectors, but deliberately does not assert a result at every odd order. Second, the Bernoulli transport filter is combined with source-independent prime-order and prime-plus-two obstructions and with explicit exceptional-order finite-ring compatibility identities. Third, at order 65 the argument eliminates all high-index sources by a uniform 5-adic coefficient-valuation bound and obtains a nonzero *function-level* Frobenius obstruction from the remaining 1,013 low-source monomials.

The Bernoulli arithmetic, finite-field elimination and p-adic methods are standard tools, not claimed as individually new. The contribution claimed here is the particular source-uniform finite-range exclusion and its specified combination of exact symbolic and computer-assisted certificates within the formal stationary CCR model. No priority claim against independently developed results, or conclusion about the physical realizability of arbitrary source vectors, is made.

3 Exact generating representation

Set $G(t,X) = \sum_{j\geq 0} F_j(X)t^j$ and write $G(t,X) = H(t)e^{Xf(t)}$. Define $Y = e^f$. Directly summing (1) and comparing the coefficients of X and the constant term gives

$$tf'(t) + Y(t) - 1 = (P+1)t + P(P+1)t^2/Y(t), \tag{2}$$

$$(\log H)'(t) = -P - P(2P+1)t/Y(t) + P\sum_{s\geq 3}(-1)^s \lambda_s t^{s-1}/Y(t)^{s-1}. \tag{3}$$

They uniquely determine $f \in t\mathbb{Q}[P][[t]]$ and $H \in 1 + t\mathbb{Q}[P,\lambda][[t]]$. Writing

$$a = \frac{P+1}{2}, \qquad d^2 = \frac{(P+1)(5P+1)}{4}$$

solves (2) as a *formal* series:

$$Y(t) = at + dt\,\coth(dt) = 1 + at + \sum_{m\geq 1} \frac{2^{2m}B_{2m}}{(2m)!} d^{2m} t^{2m}. \tag{4}$$

Only even powers of d occur, so no quadratic extension is needed. A convenient exact consequence is

$$[X^k]F_n(X) = [t^n]\; H(t)\frac{f(t)^k}{k!}. \tag{5}$$

This is a source-sensitive Sheffer representation, not an identification of the entire family with classical source-free Nörlund polynomials.

4 Bernoulli transport condition and auxiliary-prime sieves

4.1 A necessary condition at the transport prime

Proposition 4.1 (Transport-prime Bernoulli filter)**.** *Let $n \geq 5$ be odd, $P > 2n$ be prime, and $\lambda_3, \ldots, \lambda_n \in \mathbb{Z}$. If $X^3 \mid F_n(X)$ then*

$$P \mid \mathrm{num}(B_{n-1}), \tag{6}$$

where Bernoulli numbers are reduced rational fractions.

Proof. The inverse T_j^{-1} only divides by $j, j+1, \ldots, 2j$ in the degrees of interest; these are P-units for $j \leq n$ and $P > 2n$. By induction the coefficients of F_j are P-integral. Mod P the source terms and the extra transport corrections vanish, leaving the ordinary Nörlund–Stirling shadow

$$S_j(X) = [t^j]\left(\frac{t}{1-e^{-t}}\right)^X.$$

For odd $n \geq 5$, $[X^0]S_n = [X^1]S_n = 0$ and

$$[X^2]S_n = -\frac{B_{n-1}}{2(n-1)(n-1)!}. \tag{7}$$

These rational denominators are P-units because $P > 2n$. Thus $[X^2]F_n = 0$ implies $B_{n-1} = 0$ modulo P, proving (6). □

Remark 4.2. *For P-adic* sources*, the corresponding condition is not an obstruction at the same prime: the final three source variables act via an affine triangular P-adic unit matrix on the three lowest jets. Consequently Bernoulli irregularity permits a uniquely tuned P-adic terminal triple after earlier sources are fixed. This does not imply existence of an ordinary-integer terminal triple, because integrality at other primes must also hold. The finite-range theorem exploits exactly this distinction.*

4.2 Prime-order source-independent residues

Let $\ell \geq 5$ be an odd prime, and put $a = (P+1)/2$ and $D = (P+1)(5P+1)/4$. For $c_j = [X^j]F_\ell(X)$, a formal Bernoulli-pole computation using (4), (3), and von Staudt–Clausen gives

$$\ell c_2 \equiv aD^{(\ell-1)/2}, \qquad \ell c_1 \equiv a^\ell - (a+P)D^{(\ell-1)/2} \pmod{\ell}. \tag{8}$$

Both residues hold for *all* ℓ-integral source vectors. The apparent products ℓc_j are defined by first taking rational coefficients: the pole is at most simple at ℓ. The coefficient of $t^{\ell-1}$ of Y has residue $\ell[t^{\ell-1}]Y = D^{(\ell-1)/2}$. The unique pole terms in f_ℓ and $H_1 f_{\ell-1}$ then yield (8); all remaining contributions are ℓ-integral.

Proposition 4.3 (Prime-order obstruction)**.** *For any odd prime $\ell \geq 5$, any integer P, and integer sources,*

$$X^3 \mid F_\ell(X) \quad \Longrightarrow \quad P \equiv -1 \pmod{\ell}.$$

Proof. Eliminating $D^{(\ell-1)/2}$ from (8) yields

$$\ell\big((P+1)c_1 + (3P+1)c_2\big) \equiv \frac{(P+1)^2}{2} \pmod{\ell}.$$

If both c_1 and c_2 vanish, the right-hand side vanishes. Since ℓ is odd, this forces $P \equiv -1$ modulo ℓ. □

4.3 Orders two above a prime

Let $n = \ell + 2$, $\ell \geq 5$ an odd prime, and set

$$a = \frac{P+1}{2}, \quad d^2 = \frac{(P+1)(5P+1)}{4}, \quad b = \frac{(P+1)(7P-1)}{24}, \quad \Delta = (d^2)^{(\ell-1)/2}.$$

Define the finite-field polynomial

$$\begin{aligned} I_\ell(P) = -\frac{(P+1)^2}{1152}\Big[&(162P^2+100P+2)a^\ell \\ &+ (27P^3 - 269P^2 - 171P - 3)\Delta\Big] \quad \in \mathbb{F}_\ell. \end{aligned} \tag{9}$$

Proposition 4.4 (Prime-plus-two obstruction)**.** *For $n = \ell+2$ and arbitrary ℓ-integral sources,*

$$X^3 \mid F_n(X) \quad \Longrightarrow \quad I_\ell(P) = 0.$$

Proof. The Sheffer representation in Section 2 gives the exact prime-pole coefficient calculation. Write $H = \sum h_j t^j$, $f = \sum f_j t^j$, $y_2 = d^2/3$, $b = y_2 - a^2/2$, and $\Theta = -d^2\Delta/3$. Von Staudt–Clausen and Kummer congruences give

$$\ell[t^{\ell-1}]Y = \Delta, \qquad \ell[t^{\ell+1}]Y = \Theta \pmod{\ell}.$$

Every other relevant coefficient of Y is ℓ-integral, and two pole factors cannot enter a coefficient below order $\ell+2$, since $2(\ell-1) > \ell+2$ for $\ell \geq 5$. Set, in $\mathbb{F}_\ell$,

$$\begin{aligned} E &= a^\ell - a\Delta, & U &= \Delta(a^2 - y_2) + \Theta, \\ V &= \Delta(-a^3 + 2ay_2) - a\Theta, & s &= a^3/3 - ay_2. \end{aligned}$$

With $q = \ell h_\ell$, direct extraction from $H' = H(\log H)'$ gives

$$\begin{aligned} \ell h_{\ell+1} &= -Pq + P(2P+1)\Delta, \\ \ell h_{\ell+2} &= -Paq + P\big(\lambda_3 - (P+a)(2P+1)\big)\Delta. \end{aligned}$$

These expressions include *all* possible source contributions to the pole residues, not just a chosen source specialization. Using $h_1 = -P$, $h_2 = -Pa$ and $h_3 = (P^2a + P^2(2P+1) + P(2P+1)a - P\lambda_3)/3$, the three scaled terminal jets satisfy

$$\begin{aligned} \ell c_0 &= \ell h_{\ell+2}, \\ \ell c_1 &= V + h_1 U + h_2 E + h_3 \Delta + bq + a\ell h_{\ell+1}, \qquad (\mathrm{mod}\ell). \\ \ell c_2 &= aU + bE + s\Delta + h_1(aE + b\Delta) + h_2 a\Delta + (a^2/2)q, \end{aligned}$$

The linear combination $\ell[(a^2/6)c_0 + (a^2/2)c_1 + (4Pa/3 - b)c_2]$ cancels both q and λ_3. Substituting the displayed values of U, V, E, Θ, y_2 and simplifying gives exactly $I_\ell(P)$ in (9). Because 2 and 3 are ℓ-units, this is valid for all ℓ-integral sources. If $c_0 = c_1 = c_2 = 0$, then necessarily $I_\ell(P) = 0$. □

5 Certified order reductions below 57

A finite-order theorem demands a *complete* enumeration of the eligible transport primes after Proposition 4.1, with proof of primality and a source-independent obstruction for every survivor. The first 26 odd orders divide naturally into three classes:

Class	Orders	Certificate family
Odd-prime orders	$5, 11, 17, 23, 29, 37, 41, 47, 53$	Bernoulli + prime-order sieve
Prime-plus-two orders	$7, 9, 13, 15, 19, 21, 25, 31, 33, 39, 43, 45, 49, 55$	Exact shifted-order sieve and exceptional auxiliary-prime test
Composite exceptions	$27, 35, 51$	Exact finite-ring polynomial identities below

For the prime-plus-two class, the exact finite-field census checks (9) at every eligible Bernoulli transport prime. Its single zero case is $(n, P) = (21, 283)$, which is removed by a separate source-independent 5-adic coefficient congruence: for the specified integer denominator D_{21},

$$D_{21}[X^2]F_{21}(X) \equiv 2(P+1)^{11} \pmod 5;$$

at $P = 283$ this is nonzero. The remaining three composite orders are settled by:

1. $n = 27$: a second-level 13-adic polynomial ideal identity valid for all source residues.
2. $n = 35$: a second-level 17-adic polynomial ideal identity valid for all source residues.
3. $n = 51$: a complete 17-adic two-coordinate nonsquare criterion, with zero common residue roots.

The decisive finite-ring identities for these orders are recorded below; their source-polynomial coefficients are obtained by exact rational arithmetic over all bounded-weight source monomials, not by sampling integer-source vectors.

5.1 Explicit finite-ring certificates for the three composite exceptions

The following equations describe the finite checks used in the three remaining composite orders. The preliminary exact rational expansion includes *every* source monomial of admissible weight; the formulas below are the resulting small finite-field certificates, not specializations obtained by setting higher sources to zero.

Order 27. The Bernoulli filter leaves only $P = 657931$. For arbitrary early sources $z = (\lambda_3, \dots, \lambda_{24})$, let $R_{25}(z), R_{26}(z), R_{27}(z)$ be the uniquely determined *rational* terminal sources making $c_2 = c_1 = c_0 = 0$. Their exact triangular diagonal coefficients are $-P(P+1)^2/(8 \cdot 25)$, $P(P+1)/(2 \cdot 26)$ and $-P/27$. Coefficientwise exact expansion through weight 27 involves 1,049 early-source monomials. Over $\mathbb{F}_{13}$ set

$$A = \overline{13^2 R_{25}}, \qquad B = \overline{13^2 R_{27}}, \qquad T = \overline{13(R_{26} - 2R_{25})}.$$

The coefficientwise integrality calculation yields the *polynomial ideal identity* $T = 1 + Q_A A + Q_B B$, with the following explicit affine polynomials (all coefficients reduced modulo 13, $x_j = \bar{\lambda}_j$):

$$\begin{aligned}
A &= 3 + 2x_3 + 6x_4 + 9x_5 + 12x_6 + 10x_7 + 3x_8 + 8x_9 + 3x_{10} + 7x_{11} + 2x_{12} + 12x_{13},\\
B &= 3x_3 + 4x_4 + 6x_5 + 8x_6 + 11x_7 + 2x_8 + x_9 + 2x_{10} + 9x_{11} + 10x_{12} + 8x_{13},\\
Q_A &= -5 + 4x_3 + 5x_4 + x_5 - 3x_6 + 4x_7 - 4x_8 - 2x_9 - 4x_{10} - 5x_{11} + 6x_{12} - 3x_{13},\\
Q_B &= -3 - 6x_3 + 3x_4 - 2x_5 + 6x_6 + 5x_7 - 5x_8 + 4x_9 - 5x_{10} - 3x_{11} + x_{12} + 6x_{13}.
\end{aligned}$$

An integral terminal triple forces $A = B = T = 0$ and contradicts $T = 1 + Q_A A + Q_B B$. The algebraic multiplication of the displayed certificate is a finite check; deriving T from the rational CCR source expansion remains the computer-assisted coefficient step.

Order 35. The Bernoulli filter leaves only $P = 151628697551$. Let R_{33}, R_{34}, R_{35} denote the rational terminal-source polynomials, and set $A = \overline{17^2 R_{33}}$, $B = \overline{17^2 R_{35}}$, $T = \overline{17(R_{34} - 2R_{33})}$. The coefficient audit includes 4,737 source monomials. Write $\mathbf{x} = (x_3, \dots, x_{17})$ over $\mathbb{F}_{17}$; the coefficient of x_9 is zero. The exact reduction is

$$\begin{aligned}
A &= 7 + (7, 7, 16, 11, 2, 16, 0, 8, 8, 12, 16, 5, 13, 7, 5) \cdot \mathbf{x},\\
B &= 15 + (9, 2, 7, 8, 3, 7, 0, 12, 12, 1, 7, 16, 11, 2, 16) \cdot \mathbf{x},\\
Q_A &= 6 + (4, 3, 2, 12, 13, 2, 0, 1, 1, 10, 2, 7, 8, 3, 7) \cdot \mathbf{x},\\
Q_B &= 5 + (16, 13, 3, 1, 11, 3, 0, 10, 10, 15, 3, 2, 12, 13, 2) \cdot \mathbf{x}.
\end{aligned}$$

The degree-two reduced polynomial satisfies $T = 10 + Q_A A + Q_B B$ over $\mathbb{F}_{17}[\mathbf{x}]$. Consequently $A = B = T = 0$ is impossible. These short ideal identities are recorded explicitly; the exhaustive coefficient reductions used to obtain them are computer-assisted.

Order 51. The Bernoulli filter leaves exactly $P_1 = 417202699$ and $P_2 = 47464429777438199$. For $u = \bar{\lambda}_3$ and $v = \bar{\lambda}_4 + \sum_{s=5}^{17} b_s \bar{\lambda}_s$, the vectors $(b_5, \dots, b_{17})$ are respectively

$$\begin{aligned}
P_1 &: (10, 0, 9, 15, 11, 12, 1, 0, 6, 10, 13, 8, 12),\\
P_2 &: (3, 5, 12, 10, 15, 9, 14, 14, 15, 3, 1, 9, 1).
\end{aligned}$$

The exact 17-adic reduction of the two lowest nonconstant jets, after scaling by 17^3, is

$$\begin{aligned}
P_1: \quad & E_1 = -2u^2 + 8uv + 4u - 8v^2 - 5v - 6,\\
& E_2 = -7u^2 - 6uv - 7u + 6v^2 - 2,\\
P_2: \quad & E_1 = 3u^2 - 4uv + 5u + 7v^2 + 2v - 3,\\
& E_2 = 8u^2 - 5uv - 6u - 4v^2 - 2v + 5.
\end{aligned}$$

All 66,418 possible weighted source monomials are included in the finite coefficient audit for each prime; none of the omitted higher-source terms contributes modulo 17 after scaling. Lexicographic elimination in $\mathbb{F}_{17}[u, v]$ yields

$$\begin{array}{c|l} P_1 & (u + 2v + 8,\ v^2 + 6v + 2), \\ P_2 & (u + 8v - 8,\ v^2 - 8v - 6). \end{array}$$

The respective discriminants are 11 and 3 modulo 17, neither a square. Thus these pairs have no common $\mathbb{F}_{17}$-point, eliminating all integral sources.

6 Exceptional orders 57 through 75

6.1 Order 57

At $n = 57$, the exact Bernoulli numerator is

$$|\mathrm{num}(B_{56})| = 7 \cdot 113161 \cdot 163979 \cdot 19088082706840550550313.$$

The largest eligible factor is removed by a source-uniform 19-adic nonsquare obstruction computed from the exact source-polynomial jets. For the remaining two factors, let

$$E_i = \overline{19^3[X^i]F_{57}} \in \mathbb{F}_{19}[\lambda_3, \ldots, \lambda_{57}], \qquad i = 0, 1, 2.$$

Exact coefficient enumeration proves that each E_i depends only on two effective linear source coordinates $U = \overline{\lambda_3}$ and $V = \overline{\lambda_4} + \sum_{s=5}^{19} b_s \overline{\lambda_s}$, with b_s depending on the transport prime. In index order $s = 5, \ldots, 19$, the two coefficient vectors are, respectively,

$$\begin{aligned} P = 113161: &\quad (15, 11, 2, 17, 1, 12, 12, 17, 9, 3, 10, 13, 10, 2, 1), \\ P = 163979: &\quad (0, 15, 8, 0, 6, 7, 0, 10, 18, 0, 4, 11, 0, 13, 12). \end{aligned}$$

For $P = 113161$, elimination of $E_1 = E_2 = 0$ gives

$$U - 4V + 1 = 0, \qquad V^2 - 6V - 7 = 0,$$

whereas reduction of E_0 by these two equations yields $V - 9$. At $V = 9$ the displayed quadratic equals 1 modulo 19. For $P = 163979$ the corresponding equations are

$$U + 8V + 1 = 0, \qquad V^2 - 4V - 7 = 0,$$

and the reduction of E_0 is $-V - 4$; its zero $V = 15$ leaves quadratic residue $6 \neq 0$. Thus the ideal (E_0, E_1, E_2) contains 1 in both cases. The source-coordinate reduction is an *exact coefficientwise identity*, derived by an exhaustive finite-ring coefficient reduction, not an assumption that higher sources may simply be set to zero.

6.2 Orders 59 through 69

At $n \in \{59, 61, 63, 67, 69\}$ the Bernoulli admissibility sieve followed by Proposition 4.3 or 4.4 has been checked against every certified eligible transport prime by exact rational and modular arithmetic. The remaining order $n = 65$ is qualitatively different.

Theorem 6.1 (Source-uniform 5-adic obstruction at $n = 65$)**.** *For every prime $P > 130$ and every integral source vector, $X^3 \nmid F_{65}(X)$.*

Proof. The Bernoulli numerator of B_{64} has exactly two eligible transport prime factors:

$$P_1 = 1226592271, \qquad P_2 = 870573153545221791849896997917 27.$$

Put $c_i = [X^i]F_{65}(X)$ and $E_1 = \overline{5^{17}c_1}$, $E_2 = \overline{5^{18}c_2}$ in $\mathbb{F}_5$. A complete min-plus source-weight bound for the coefficients of the Sheffer expansion proves that every monomial involving some λ_s with $s \geq 6$ vanishes from both E_1 and E_2. This coefficientwise bound holds *uniformly* for all integer sources, including arbitrary high-index ones. The exact min-plus convolution yields the following lower bounds on 5-adic coefficient valuations:

	$\lambda_3, \lambda_4, \lambda_5$ only	at least one λ_s, $s \geq 6$
c_0	-16	-14
c_1	-17	-15
c_2	-18	-16

In particular $5^{17}c_1$ and $5^{18}c_2$ are integral coefficientwise, and terms involving a source index at least six reduce to zero modulo 5.

Let $a = \bar{\lambda}_3$, $b = \bar{\lambda}_4$ and $c = \bar{\lambda}_5$. The exact surviving low-source polynomials for $P = P_1$ are

$$E_1 = 1 + 4a + 2b + 4c + 3a^5 + 4b^5 + 3c^5, \qquad E_2 = 3 + 3a^5 + 4b^5 + 3c^5,$$

and for $P = P_2$ they are

$$E_1 = b + 3c + 2b^5 + c^5, \qquad E_2 = 3 + 2b^5 + c^5.$$

Reducing modulo the *function* ideal $(a^5 - a, b^5 - b, c^5 - c)$ gives

$$E_1 - 4E_2 = \begin{cases} 4, & P = P_1, \\ 3, & P = P_2. \end{cases} \qquad \text{on } \mathbb{F}_5^3.$$

Both right-hand sides are nonzero. Thus c_1 and c_2 cannot both vanish for any integer source vector. The computer-assisted finite input is the exhaustive source-weight valuation bound and exact 1,013-monomial low-source coefficient calculation, summarized quantitatively below. These computations are not replaced by substituting finitely many source values. □

6.3 Orders 71, 73, and 75

The exact Bernoulli numerators and complete prime certificates leave six eligible transport-prime cases in these three orders. Applying Proposition 4.3 at $n = 71$ and Proposition 4.4 at $n = 73, 75$ gives the following reproducible table:

n	Eligible transport prime P	Test	Value
71	688531	P mod 71	44
71	20210499584198062453	P mod 71	12
71	3090850068576441179447	P mod 71	18
73	3112655297839	$I_{71}(P)$	10
73	18723419087606889767942264996363043575678 11	$I_{71}(P)$	68
75	92303830511408562200892091166142257261319750 7651	$I_{73}(P)$	35

At order 71 the required residue is 70; at orders 73 and 75 the invariant is required to vanish. None of the six cases satisfies its necessary condition. The displayed candidates arise from exact Bernoulli fractions, complete numerator factorizations and recursive primality certificates for the corresponding factor trees. Its reproduced final assertion is "orders 71, 73, 75 globally excluded for all integer sources; PASS."

7 Completion of the finite-range proof

Proof of Theorem 1.1. For $5 \le n \le 55$ the odd-order set is partitioned by the three mutually exclusive rows in Section 4: $9 + 14 + 3 = 26$ orders, each treated by the cited Bernoulli and auxiliary-prime certificates. Order 57 is covered by the 19-adic three-jet computation of Section 5. Of the six odd orders $59, 61, 63, 65, 67, 69$, five are covered by the exact transport-prime sieve and order 65 by Theorem 6.1. The three remaining odd orders $71, 73, 75$ are covered by the explicit table in Section 6. Thus every one of the $26 + 1 + 6 + 3 = 36$ odd orders in the interval is excluded, with all prime transport parameters and all ordinary integer source vectors quantified universally. □

Remark 7.1 (Quantifier firewall)**.** *A source vector in $\mathbb{Q}_P$, a vector in $\mathbb{Z}_P$, and an ordinary integer vector are different objects. Local solvability at the transport prime does not contradict any theorem in this paper. The universal quantifiers in this theorem are over all transport primes and ordinary integer sources for the stated finite set of orders; no claim about physical machines follows.*

Exact verification and finite-certificate scope

The theorem combines symbolic arguments with exact, exhaustive computer-assisted finite arithmetic. The computations are part of the proof at the exceptional degrees; they are not merely randomized consistency experiments and do not replace the universal symbolic reductions to finite residue rings. Specifically:

Orders / stage	Exact finite assertion and scope
$5 \le n \le 55$	Complete Bernoulli transport-prime factorizations and prime/shifted-order congruences; the single $(21, 283)$ residual case is excluded by the 5-adic congruence displayed above.
$27, 35$	All 1,049 / 4,737 early-source monomials in the terminal-source expansion; the resulting explicit polynomial ideal identities with residues 1 / 10.
51	All 66,418 possible weighted source monomials for each eligible transport prime; exact two-coordinate reduction and the two nonsquare discriminants.
57	All 162,878 possible weighted monomials through source weight 57, and two 19-adic, three-jet unit-ideal tests; the third eligible prime is separately excluded.
65	All-index min-plus valuation bounds and all 1,013 low-source monomials for each of the two eligible primes; function-level Frobenius obstruction in $\mathbb{F}_5^3$.
$71, 73, 75$	Complete Bernoulli numerator factorization and primality checks for eligible factors, followed by six exact finite-field congruence tests in the displayed table.

The finite arithmetic uses exact rational coefficients, prime certificates and modular polynomial identities, with no floating-point tolerance. The displayed formulas record the mathematical outcomes used by the proof. The underlying full coefficient enumerations, although internally audited, are not reproduced line by line in this article, and no source-code repository or downloadable verification supplement is represented as part of this publication. A complete independent replication of those computer-assisted stages would require regenerating the stated finite expansions; the symbolic lemmas alone do not establish those finite assertions.

Conclusion

Theorem 1.1 excludes a triple zero at the origin for all 36 odd orders from 5 to 75, every eligible prime transport and all integral source choices. It separates a general Bernoulli transport

filter and auxiliary-prime criteria from the genuinely source-polynomial obstructions required at exceptional composite orders. All stated claims are about the explicitly defined stationary CCR polynomial recurrence and the specified finite range.

Statements and Declarations

Author contributions. Alp Eren Bütün: conceptualization, methodology, formal analysis, investigation, software, validation, writing—original draft, and writing—review and editing.
Funding. No external funding was received for this research.
Competing interests. The author declares no relevant financial or non-financial competing interests.
Data availability. This is a mathematical study and uses no empirical or personal dataset; the relevant definitions, statements and symbolic proofs are contained in the article. The finite-arithmetic verification outcomes used in the computer-assisted exceptional-order arguments are summarized in the proof and exact verification discussion. No supplementary code or data archive is part of this publication.
Ethics approval and consent to participate. Not applicable.
Consent for publication. Not applicable.

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